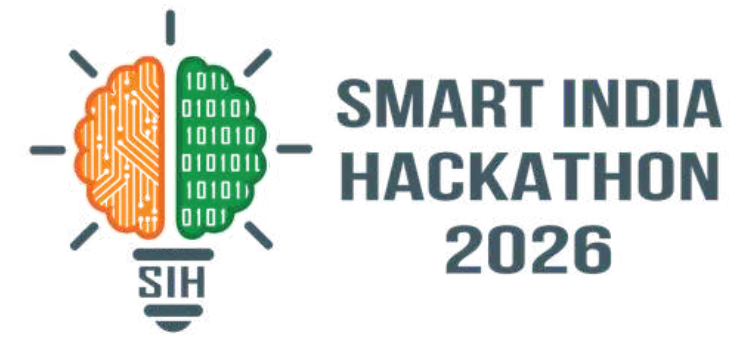
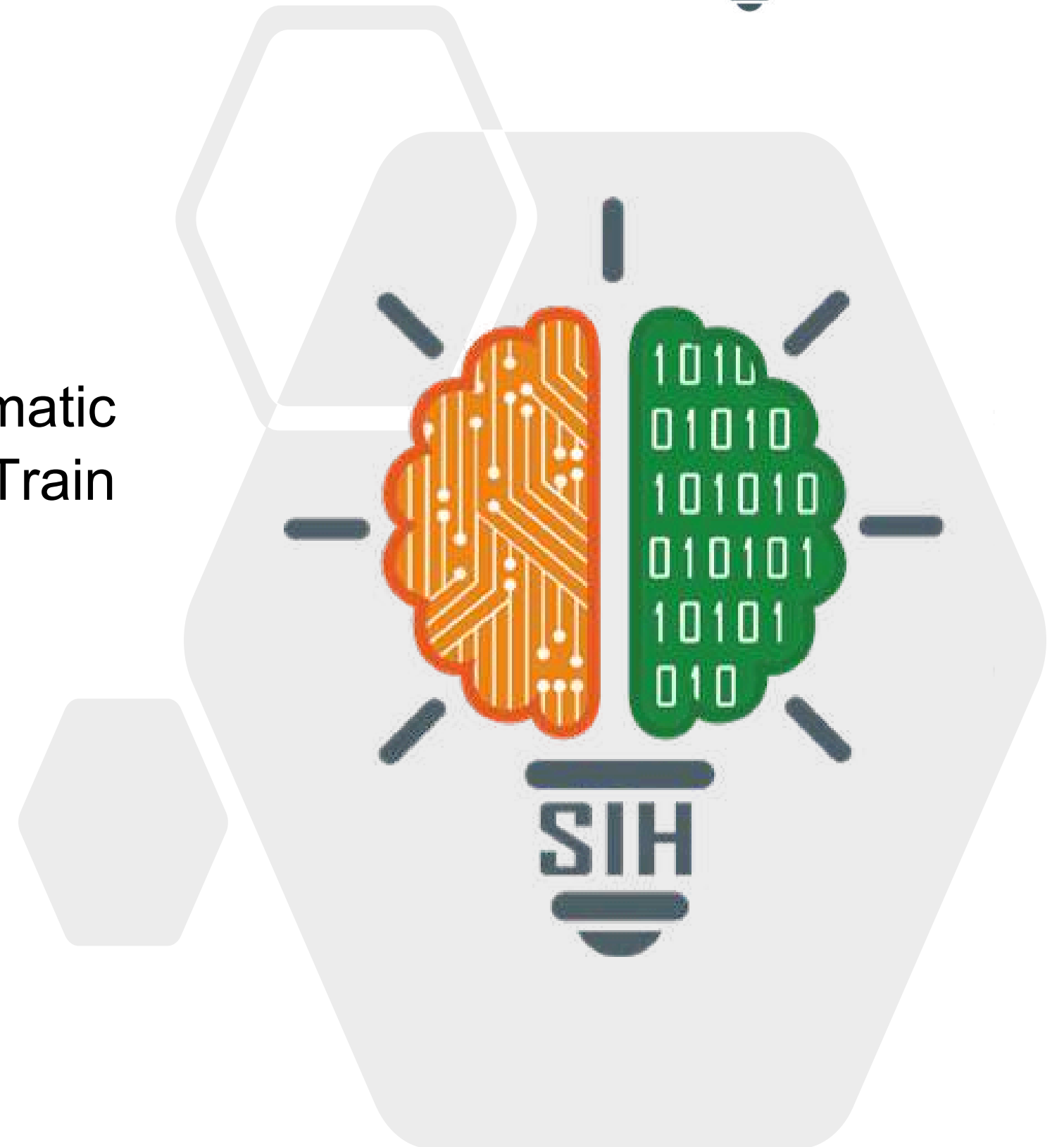


SMART INDIA HACKATHON 2026



Title Page

- **Problem Statement ID – SIH26027**
- **Problem Statement Title -** AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways
- **Theme-** Transportation and Logistics
- **PS Category -** Software
- **Team ID -** SIH26027/0214
- **Team Name -** LastCommit



PROBLEM : How can Indian Railways intelligently coordinate maintenance blocks to avoid train conflicts, repeated track closures, delays, and asset downtime?

SOLUTION in short : An intelligent railway scheduling system that analyzes train schedules, asset health, and maintenance requests to detect conflicts, find optimal maintenance windows, and coordinate work into unified Corridor Mega-Blocks.

Explanation of the Proposed Solution

Algorithmic decision-support vs manual paper planning

- **Automated Planning Platform:** Integrates maintenance requests and reduces manual coordination and trail-and-error planning.
- **Multi-Source Data Ingestion:** Integrates live Train Timetables (COA), Track Defects (TMS), Signalling (SMMS), and Overhead Equipment (TDMS).
- **AI-Based Maintenance Prioritization:** Prioritizes maintenance tasks based on criticality, urgency and impact on asset availability.
- **Multi-Horizon Rostering:** Generates high-level Monthly Maintenance Calendars and refined Weekly Block Rosters with dynamic real-time re-planning.
- **Traffic-Aware Data:** Integrates COA block availability, train timetable and goods-train forecast.

How it addresses the problem

Train-aware planning & departmental coordination

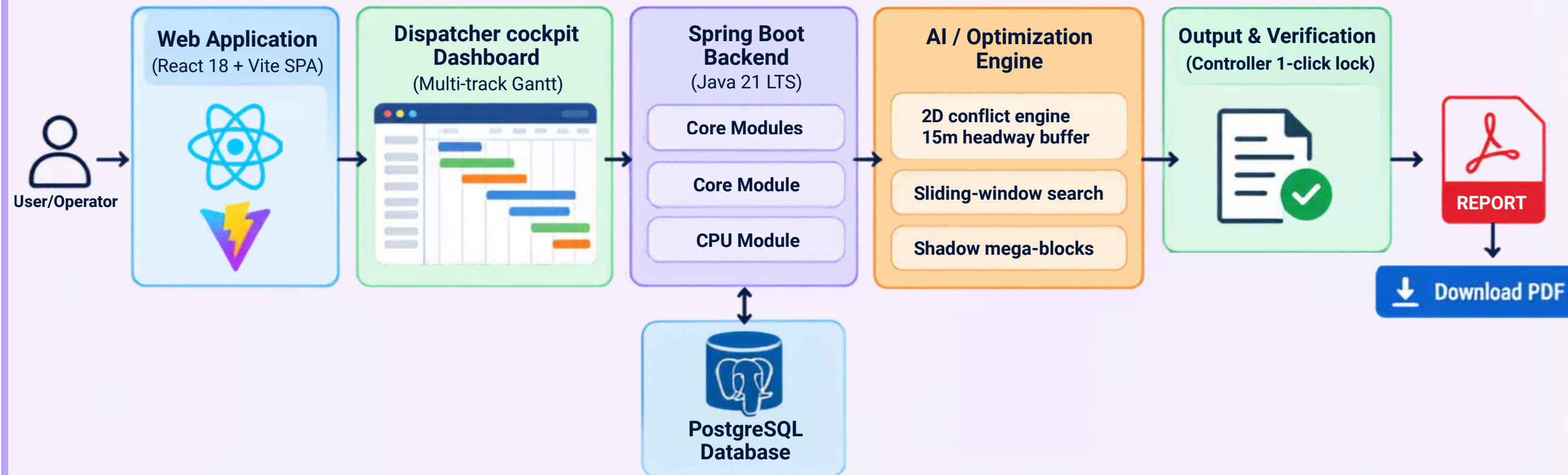
- **Safety & Operational Constraints:** Ensures recommended blocks follows required operational and safety.
- **Breaks Departmental Silos:** Bridges Civil (P-Way), Signal & Telecom (S&T), and Electrical (TRD) to eliminate repeated closures on identical corridors.
- **Multi-Tier Priority Scorer:** Weights train classes (Vande Bharat = 5.0, Express = 2.5, Freight = 1.0) to strictly penalize delays on premium passenger services.
- **Dispatcher Cockpit Visibility:** Delivers live interactive Gantt charts with 1-click AI recommendation locks and automated collision alerts for Section Controllers.

Innovation and uniqueness of the solution

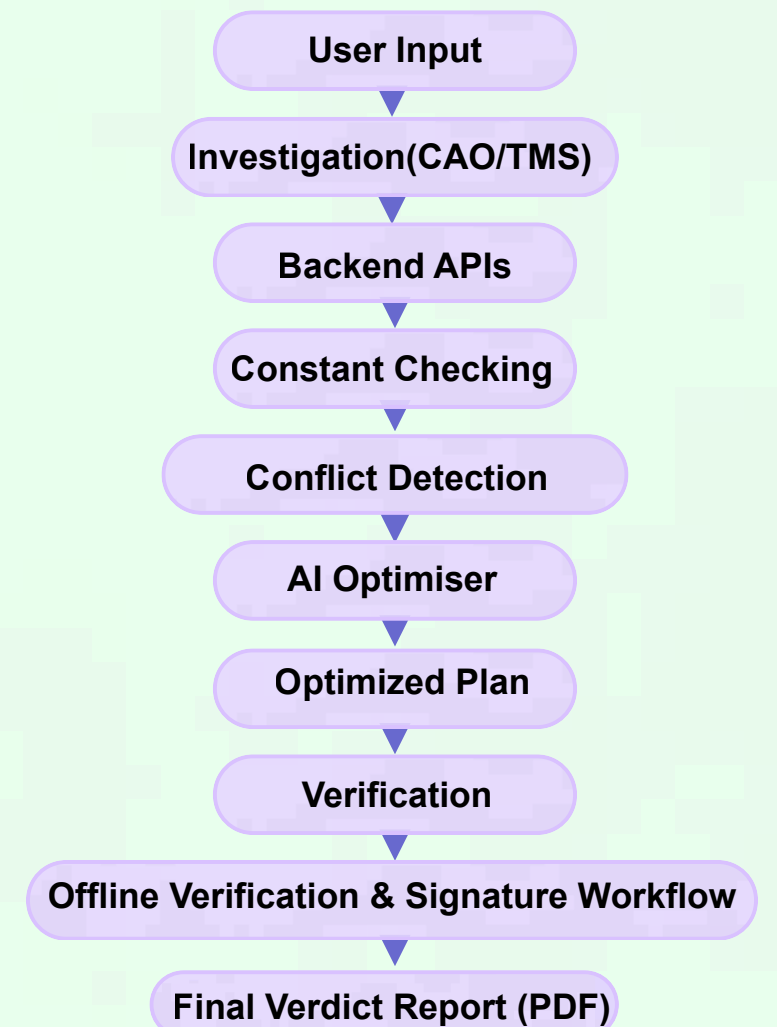
Breakthrough technical differentiators

- **"Shadow Blocking" & Mega-Blocks:** When Track A closes for rail tamping, system automatically scans degraded S&T / OHE assets (<70% health) on Track A and bundles repairs into one joint closure.
- **Sliding-Window AI Optimizer:** Heuristic 24-hr scan in 30-minute steps computing Disruption Cost to recommend Top 3 conflict-free alternative slots.
- **Explainable AI (XAI):** Gives plain-English operational reasoning for each recommendation ("Zero train conflicts, bundles 3 assets, saves 270 min").
- **Northern Railway Validation:** Tested on Delhi–Ghaziabad Quadruple Trunk Corridor (11 clashes → 0; 8.5h downtime restored; 270 min saved)
Track + OHE + S&T → Compatible works → Common Coordinated Block.

SYSTEM ARCHITECTURE



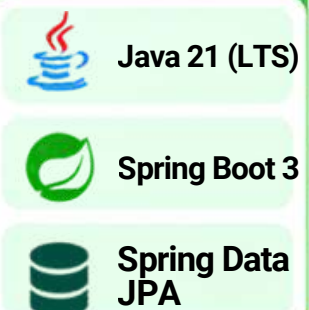
DATA FLOW



FRONTEND



BACKEND



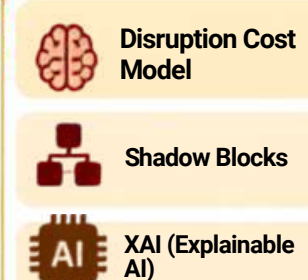
DATABASE



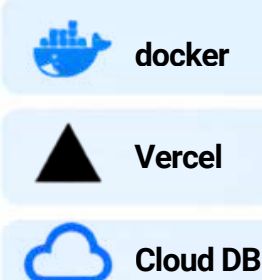
ALGORITHMS



AI / DECISION ENGINE



DEPLOYMENT



TOOLS & PDF



 Technical Feasibility (100% Proven)

Validated working full-stack prototype

- **Enterprise Architecture:** Built on enterprise Java 21, Spring Boot, and PostgreSQL; highly modular, resilient, and horizontally scalable.
- **Sub-Second Latency:** Conflict check takes <50ms; 24-hr schedule optimization finishes in <200ms, ensuring instantaneous UI feedback.
- **Authentic Dataset Proof:** Pre-seeded and verified on Northern Railway NDLS–GZB quadrupled trunk dataset (16 tracks, 16 assets, 12 trains).

**Operational Feasibility (Zero Friction)**

Seamless integration into IR workflows

- **Preserves Hierarchy:** Seamlessly overlays on existing BDMS and COA procedures without altering command structures or retraining staff.
- **Human-in-the-Loop:** Controllers retain full final override authority; AI serves as an intelligent advisory copilot, not a black-box replacement.
- **Intuitive Cockpit:** Replaces dense paper registers with a modern visual Gantt timeline showing real-time track occupation.

**Financial & Resource Viability**

High ROI & minimal infrastructure cost

- **Zero Hardware Mandate:** Pure software intelligence; runs on commodity cloud (NIC/AWS) or IR data centers with zero bespoke line-side track sensors.
- **Rapid Financial ROI:** Slashes passenger delay penalties, eliminates diesel train idling wastage, and avoids catastrophic emergency track repairs.
- **Asset Longevity:** Predictive shadow-blocking prevents neglected point machines and OHE wires from reaching catastrophic failure.

Operational Challenge / Risk	Potential Impact on Operations	Technical Mitigation Strategy	Status & Feasibility
Departmental Data Silos (TMS, SMMS, TDMS in separate legacy systems)	Incompatible formats, delayed approvals, duplicate line possessions	Unified Ingestion Adapter Layer: Standardized REST/JSON schemas, data validation, and automated staging ETL pipelines	Prototype Built & Verified
Dynamic Operational Disruptions (Unscheduled freight, sudden delays, rail fractures)	Pre-planned maintenance block clashes with delayed passenger trains	Event-Driven Dynamic Re-Planning Engine: Automated rescheduling triggered by live COA delay feeds in <200ms	Algorithm Verified in Code
Controller Adoption & Skepticism (Staff resistance to automated algorithms)	Controllers ignore recommendations and rely on manual phone calls	Explainable AI (XAI) Cockpit: Displays transparent operational rationale ('Zero conflicts, bundles 3 assets, saves 270 min')	UI Implemented & Active

ENVIRONMENTAL IMPACT

IMPACT

Reduce Train Idling - Better block coordination minimizes avoidable train detention during maintenance.

Reduce Machine Idle Time - Blocks are planned around maintenance-machine availability, reducing unnecessary waiting and running.

Reduce Unnecessary Machine Movement - Work is planned considering machine location and sequence, avoiding avoidable deployment and movement.

Improve Energy Efficiency - Fewer conflicting blocks and stop-wait cycles reduce wasted traction and operational energy.

BENEFIT

→ Lower fuel & energy consumption, reduced operational waste, and a smaller environmental footprint.

SOCIAL IMPACT

IMPACT

Fewer Passenger Delays - Maintenance windows are planned around train movement, reducing avoidable train-maintenance conflicts.

Better Train Availability - Critical passenger services face fewer unnecessary disruptions from maintenance activity.

Safer Maintenance Windows - Clear separation between train movement and maintenance work creates safer working conditions for railway staff.

Better Operational Coordination - Centralized block status and conflict information keeps controllers and maintenance teams aligned.

BENEFIT

→ More reliable journeys, safer maintenance operations, and less disruption for passengers and commuters.

ECONOMIC IMPACT

IMPACT

Higher Track Capacity Utilization - Optimized block timing makes more productive use of existing railway infrastructure.

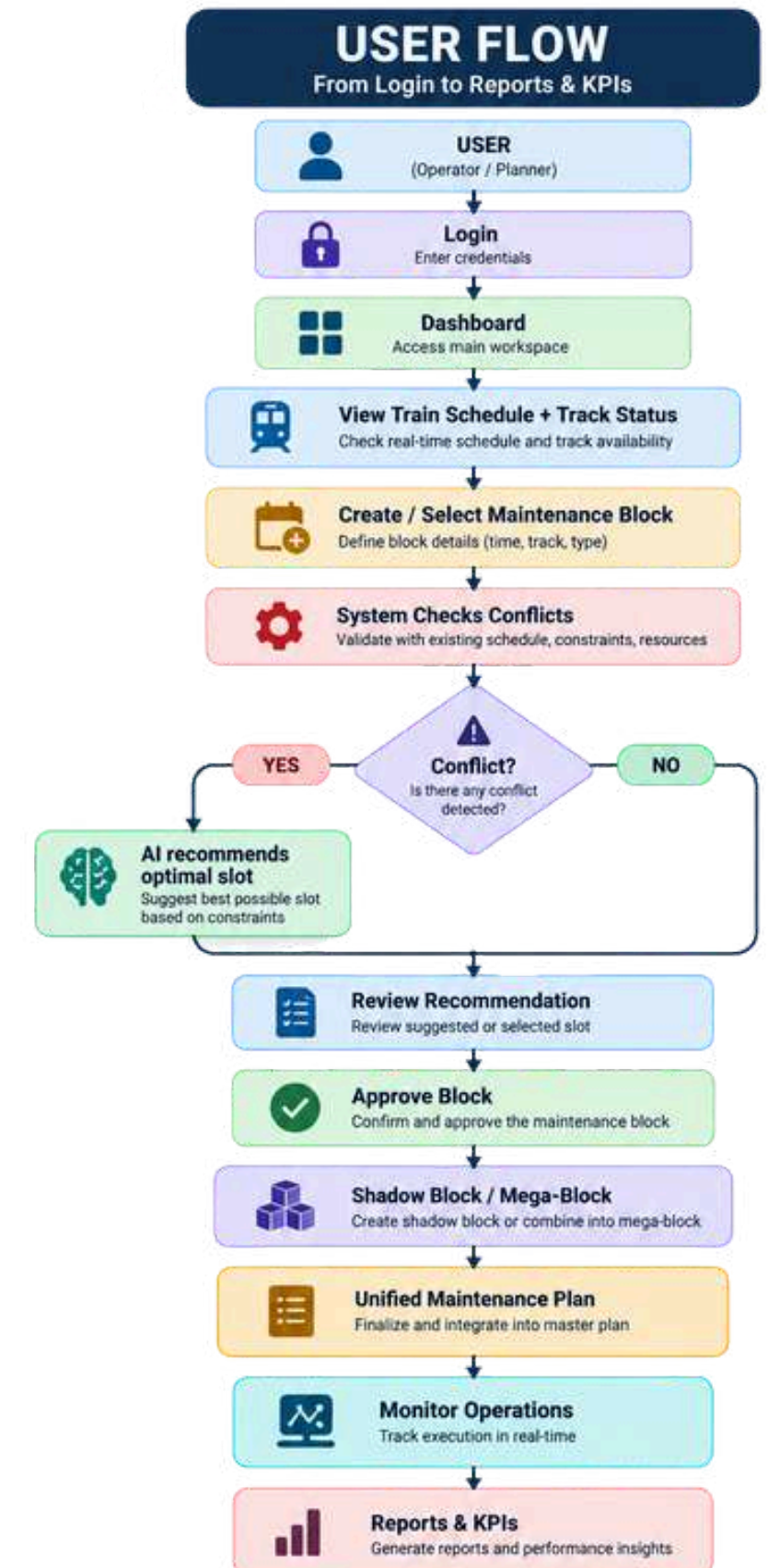
Reduced Train Detention - Fewer avoidable waits improve utilization of locomotives, coaches and wagons.

Better Maintenance Asset Utilization - Matching machines with suitable blocks reduces idle time of expensive maintenance equipment.

Lower Operational Waste - Detecting conflicting or overlapping requests before approval prevents wasted block hours and unnecessary deployment.

BENEFIT

→ Better utilization of existing assets, lower operational waste, and improved railway capacity efficiency.



Indian Railways Institutional Groundwork

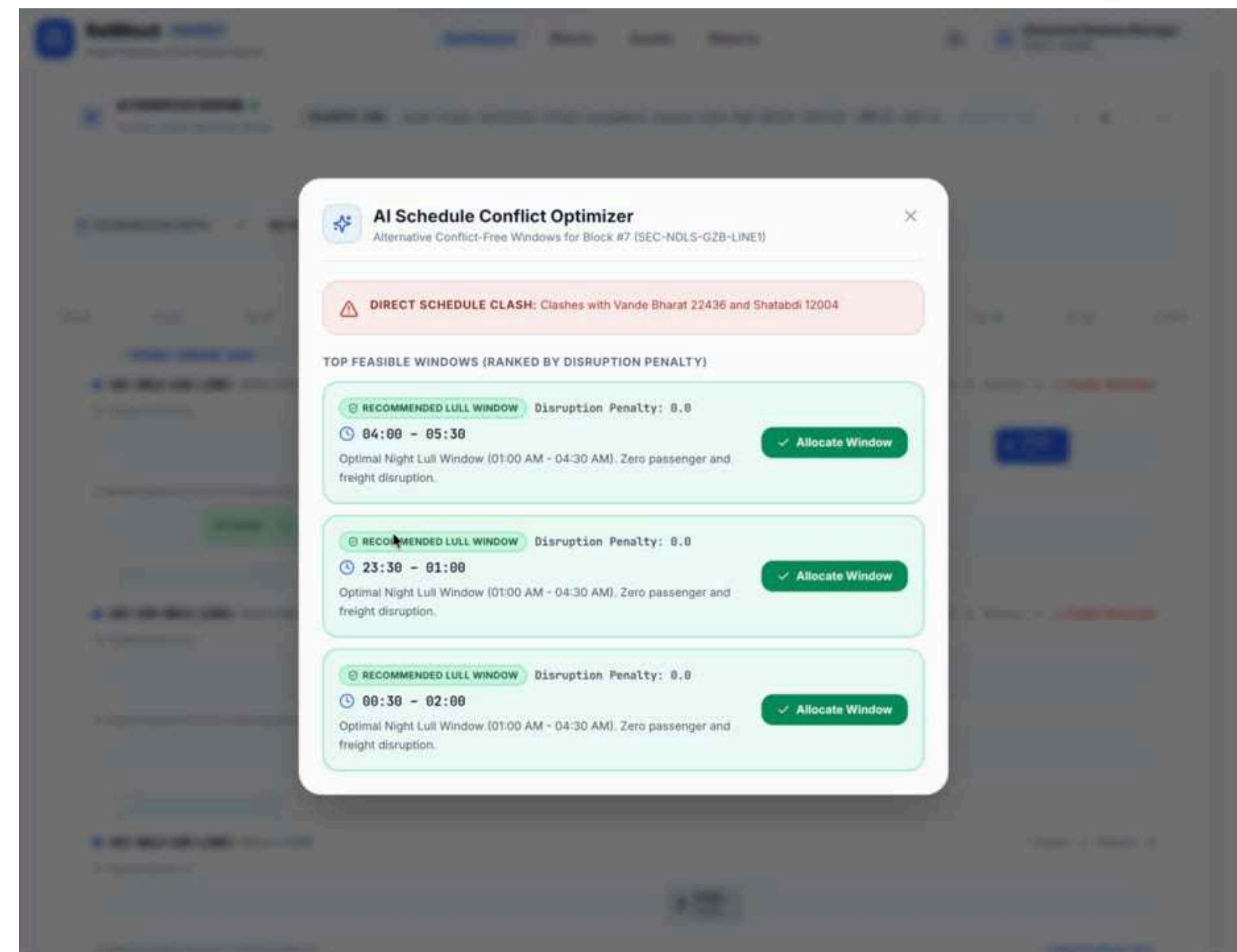
- **CAG Railway Audit (Report No. 22):** Empirical data connects uncoordinated maintenance & asset failures to 22.20% of total railway punctuality loss.
LINK: ([CAG Report No. 22 of 2021](#)).
- **CAMTECH Guidelines:** Standard Operating Procedures for Traffic & Power Blocks, 25kV OHE isolation, work permits, and machine/staff log registers
LINK: ([CAMTECH Handbook](#)).
- **IRICEN Rolling Block Research:** Studies on rolling block programs, corridor-based maintenance coordination, and AI block-window identification methodologies
LINK:([IRICEN Rolling Block Research](#)).

Industry Research & AI Initiatives

- **IndiaAI (Ministry of Railways):** Documented AI use-cases in track condition assessment, acoustic bearing monitoring, and predictive signalling maintenance frameworks.
LINK: ([IndianAI-Ministry of Railway](#)).
- **TCS Railway Asset Management Research:** AI/ML frameworks demonstrating quantifiable line-capacity benefits of bundling multi-department maintenance into consolidated access windows.
LINK: ([TCS Research](#)).
- **Indian Railways G&SR Standards:** General & Subsidiary Rules governing absolute/automatic block working, safe braking headways, and Control Office Application (COA) protocols.
LINK: ([Indian Railways G&SR-2025](#)).

Global Railway Benchmarks

- **JR East (Japan):** AI-based proactive condition monitoring, smart track inspection cars, and condition-based predictive maintenance protocols.
LINK: ([JR East](#)).
- **Deutsche Bahn (Germany):** AI-driven train dispatching and automated timetable optimization engines for ultra-dense mixed passenger/freight rail networks.
LINK: ([Deutsche Bahn](#)).
- **European Rail Joint Scheduling:** Scientific research models for simultaneous mathematical optimization of train paths and scheduled infrastructure track possession.
LINK:([Research Paper](#)).



BRIDGING THE MISSING IN INDIAN RAILWAYS

- Existing RBPMS/workflows + AI optimization → intelligent coordinated block planning.
- Our Breakthrough: An end-to-end algorithmic decision layer that transforms siloed maintenance requests into conflict-free, multi-department Mega-Blocks, strictly enforcing safety buffers and optimizing asset uptime.
- Empirical Reproducibility: Tested on authentic Northern Railway operational simulations with verified mathematical before-vs-after metrics.
LINK: [RBPMS](#).